$$\ddot{x}(t) = F(x, \dot{x}, t) = F(x, t) - \gamma \dot{x}(t)$$

$$\cancel{V(t+dt) = F(x, \dot{x}, t)}$$

$$\dot{x}(t+dt) = \dot{x}(t) + \ddot{x}\, dt = \dot{x}(t) + F(x,t)\, dt - \gamma \dot{x}(t)\, dt$$

$$\dot{x}(t+dt) = \dot{x}(t)\,(1 - \gamma\, dt) + F(x,t)\, dt$$

$$dt = \frac{h}{2}$$

$$\dot{x}\left(t + \frac{h}{2}\right) = \dot{x}(t)\left(1 - \frac{h\gamma}{2} dt\right) + \frac{h}{2} F(x,t) \qquad 2.16$$

$$\dot{x}\left(t + \frac{h}{2}\right) = \frac{x(t+h) - x(t)}{h}$$

$$x(t+h) = x(t) + h\, \dot{x}\left(t + \frac{h}{2}\right) \qquad 2.17$$

$$\cancel{F[x(t+h), t+h] =} \quad F(x,t) = 2ax - 4bx^3 + F_0 \cos(\omega t)$$

$$F[x(t+h), t+h] = 2a\, x(t+h) - 4b\, x^3(t+h) + F_0 \cos(\omega(t+h))$$

$$= 2a\, x(t) + 2h\, \dot{x}\left(t + \frac{h}{2}\right) + F_0 \cos(\omega t + \omega h) - 4b\,(x(t+h))^3$$

$$\cancel{-4b(x(t+h))^3 - x^3(t) +}$$

$$= \cancel{2a\, x(t) +} \; 2a\, x(t+h) - 4b\,(x(t+h))^3 + F_0 \cos(\omega t + \omega h)$$

$$2.18$$

$$V\left(t + \frac{h}{2}\right) = V(t+h) - \frac{h}{2} \ddot{x}(t+h)$$

$$V\left(t + \frac{h}{2}\right) = V(t+h) - \frac{h}{2} F[x(t+h), t+h] + \frac{h}{2} \gamma V(t+h)$$

$$V\left(t + \frac{h}{2}\right) = \left(1 + \frac{h}{2}\gamma\right) V(t+h) - \frac{h}{2} F[x(t+h), t+h]$$

$$\left(1 + \frac{h}{2}\gamma\right) V(t+h) = V\left(t + \frac{h}{2}\right) + \frac{h}{2} F[x(t+h), t+h]$$

$$V(t+h) = \frac{V\left(t + \frac{h}{2}\right) + \frac{h}{2} F[x(t+h), t+h]}{\left(1 + \frac{h}{2}\gamma\right)} \qquad 2.19$$